

FIITJEE

ALL INDIA TEST SERIES

FULL TEST – XV

JEE (Main)-2022

TEST DATE: 15-06-2022

ANSWERS, HINTS & SOLUTIONS

Physics

PART – A

SECTION – A

1. B

Sol. Given that, $K = as^2$ or $\frac{1}{2}mv^2 = as^2$

$$\text{or } mv^2 = 2as^2 \quad \dots(i)$$

Differentiating w.r.t time t, m. $2v \frac{dv}{dt} = 2a \cdot 2s \cdot \frac{ds}{dt}$

$$\text{But, } \frac{ds}{dt} = v$$

$$\therefore 2m \frac{dv}{dt} = 4as \quad \text{or } m \frac{dv}{dt} = 2as$$

Now, $m \frac{dv}{dt}$ = tangential force = F_t

$$\therefore F_t = 2as \quad \dots(ii)$$

$$\text{Centripetal force} = F_r = \frac{mv^2}{R} = \frac{2as^2}{R} \quad \dots(iii)$$

$$\therefore F = \sqrt{F_t^2 + F_r^2} = \sqrt{(2as)^2 + \left(\frac{2as^2}{R}\right)^2}$$

$$= 2as \sqrt{1 + \frac{s^2}{R^2}}$$

2. A

Sol. This error occurs in all the instruments which utilize screw or nut-bolt system

3. A

$$\text{Sol. } \frac{50 - 45}{5} = K \left(\frac{50 + 45}{2} - \theta_s \right)$$

$$\frac{45 - 40}{8} = K \left(\frac{45 + 40}{2} - \theta_s \right)$$

$$\text{Hence } \theta_s = 34^\circ \text{C}$$

4. A

 Sol. Here, $v = 1.5 \text{ ms}^{-1}$

$$\frac{dm}{dt} = 5 \text{ kgms}^{-1}, F = \frac{dm}{dt} \times v = 5 \times 1.5 = 7.5 \text{ N}$$

$$P = F \times v = 7.5 \times 1.5 = 11.25 \text{ W}$$

5. A

$$\text{Sol. } \lambda = \frac{h}{\sqrt{2meV}} \text{ and } \lambda_0 = \frac{hc}{eV} = \frac{2mc\lambda^2}{h}$$

6. B

Sol. As, impulse received;

$$\text{As, } J = 9 \times 0.02 = m(V - 0)$$

$$V = \frac{0.18}{3} = 0.06 \text{ m/sec}$$

$$\text{Velocity after 3 sec} \Rightarrow V' = 0.06 \text{ m/sec}$$

$$3 \text{ sec} \Rightarrow V' = 0.06 \text{ m/sec}$$

Now impulse received by second force

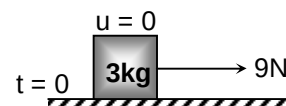
$$J' = -9 \times 0.01 = m(V'' - V')$$

$$V'' = -0.03 + 0.06 = 0.03 \text{ m/sec}$$

Alternate solution

$$J = 9(0.02 - 0.01) = m(V'' - 0)$$

$$V'' = 3 \text{ cm/s}$$



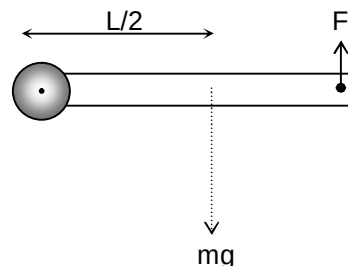
7. B

Sol. Taking torque about hinge:

$$Mg \frac{L}{2} = FL$$

$$mg \frac{L}{2} = 2 \left(\frac{mv}{10} \right) 10L$$

$$v = 2.5 \text{ m/sec}$$



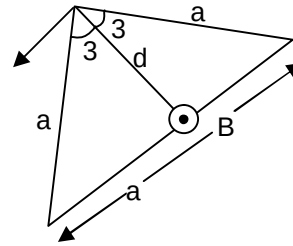
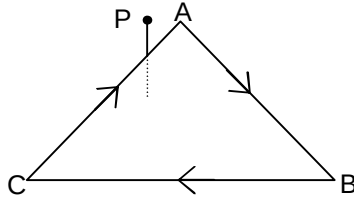
8. C

$$\text{Sol. } e = \frac{-d\phi}{dt}$$

9.

A

Sol. At P only vertical components of magnetic field will add up due to each wire.



$$B_{\text{net}} = 3 \times \frac{\mu_0 L}{4\pi d} \times \frac{1}{3}$$

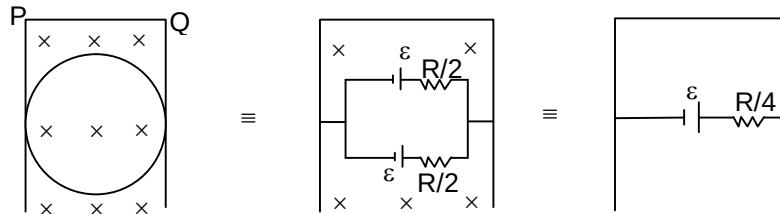
$$\text{where } d = \frac{\sqrt{3}a}{2}$$

10.

D

Sol. $\varepsilon = BIV = B(2r)V$

$$I = \frac{\varepsilon}{R/4} = \frac{8BrV}{R}$$

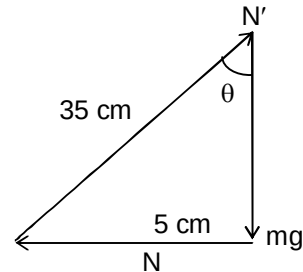


11.

B

Sol. $\tan \theta = \frac{N}{mg}$

$$\Rightarrow \frac{5}{\sqrt{(35)^2 - (5)^2}} = \frac{N}{mg}$$



12.

A

Sol. Due to $2i$ (w_1)

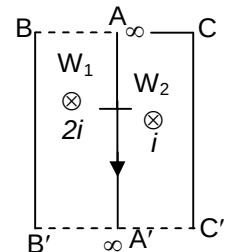
$$\int_A^{A'} \vec{B} \cdot d\vec{l} + \int_{A'}^{B'} \vec{B} \cdot d\vec{l} + \int_{B'}^B \vec{B} \cdot d\vec{l} + \int_B^A \vec{B} \cdot d\vec{l} = \mu_0(2i)$$

$$A' \int_A^{A'} \vec{B} \cdot d\vec{l} = \int_{B'}^B \vec{B} \cdot d\vec{l} \quad (\text{By symmetry})$$

$$\int_A^{A'} \vec{B} \cdot d\vec{l} = \frac{\mu_0 2i}{2} = \mu_0 i$$

Similarly due to $i(w_2)$ $A' \int_A^{A'} \vec{B} \cdot d\vec{l} = \frac{\mu_0 i}{2}$ in opposite sense hence

$$\int_A^{A'} \vec{B} \cdot d\vec{l} (\text{Total}) = \mu_0 i - \frac{\mu_0 i}{2} = \frac{\mu_0 i}{2}$$



13. B
Sol. Conceptual

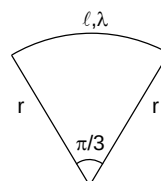
14. C

Sol. Length of the arc $= r\theta = \frac{r\pi}{3}$

Charge on the arc $= \frac{r\pi}{3} \times \lambda$

$\therefore$ Potential at center $= \frac{kq}{r}$

$= \frac{1}{4\pi\epsilon_0} \times \frac{r\pi\lambda}{3r} = \frac{\lambda}{12\epsilon_0}$



15. C

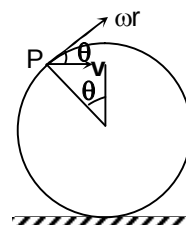
Sol. The temperature of these objects being identical the cooling rate will depend upon surface area. The circular plate has the largest area and hence will cool faster than cube and sphere, Sphere having less surface area than a cube will cool at the lower rate

16. B

Sol. $v_p = \sqrt{v^2 + v^2 + 2v^2 \cos \theta}$

($\omega r = v$)

$= \sqrt{2v^2(1 + \cos \theta)} = \frac{2v \cos \theta}{2}$

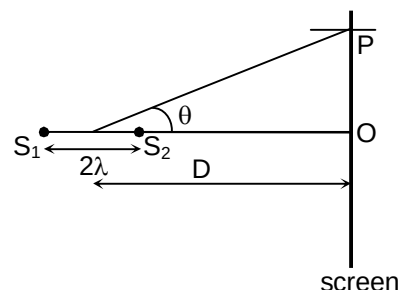


17. C

Sol. Referring to the figure, the path difference between the two waves starting from S_1 and S_2 turns out to be $(2\lambda \cos \theta) = n\lambda$ where n is taken as 1 to get the point of maximum intensity which is the same as a point O. Therefore, the above relation gives

$\cos \theta = 1/2$ so that $\theta = 60^\circ$

and $\tan \theta = PO / D = \sqrt{3}$, giving $PO = D\sqrt{3}$



18. A

Sol.

A	B	Y	Y ₁	Y ₂	Y
0	0	1	1	1	0
0	1	1	1	0	1
1	0	1	0	1	1
1	1	0	1	1	0

19. C

Sol. Radioactivity does not depend on physical and chemical factors

20. C

Sol. $r = \frac{mv \sin \theta}{Bq}$

$$T = \frac{2\pi m}{Bq}$$

$$x = V \cos \theta \times T$$

SECTION – B

21. 00040.00

Sol. Here; $D = 25$ cm, angular magnification (M) = 10

$$\text{As } M = \frac{D}{f}, f = \frac{D}{M} = \frac{25 \text{ cm}}{10} = 2.5 \text{ cm}$$

$$\text{Power of the microscope, } P = \frac{100}{2.5 \text{ cm}} = 40 \text{ diopters}$$

22. 00004.00

$$\text{Sol. } 8 = \frac{1}{\cos^3 \alpha} \Rightarrow \cos \alpha = \frac{1}{2}$$

$$\Rightarrow \alpha = 60^\circ = 15\theta \Rightarrow \theta = 4$$

23. 00003.00

$$\text{Sol. } F_1 = mg \sin \theta + \mu mg \cos \theta$$

$$F_2 = mg \sin \theta - \mu mg \cos \theta$$

$$\therefore \frac{F_1}{F_2} = \frac{\tan \theta + \mu}{\tan \theta - \mu} = 3$$

24. 00003.00

$$\text{Sol. As } 3v = (m+3)\left(\frac{v}{3}\right), m = 6 \text{ kg}$$

$$\frac{k_1}{k_2} = \frac{\frac{1}{2} \times 3 \times v^2}{\frac{1}{2} \times 9 \times \frac{v^2}{9}} = 3$$

25. 00016.00

$$\text{Sol. } \frac{1}{2} \cdot \left(\frac{C_1}{n_1}\right) (4V)^2 = \frac{1}{2} (n_2 C_2) V^2$$

$$\therefore C_2 = \frac{16C_1}{n_1 n_2}$$

26. 00003.00

$$\text{Sol. } v = \pi r^2 l$$

$$\% \text{ error in } V = \% 2 \times \text{error in } r + \% 1 \times \text{error in } l$$

$$= \left(\frac{2 \times 0.01}{2} + \frac{0.1}{5} \right) \times 100 = 3\%$$

27. 00005.00

Sol. Let $A \text{ cm}^2$ be the surface area of the bottom of the tank which is in contact with the furnace.
Volume of water evaporating in a time t (9 minute = 540 s)
= area $\times$ thickness of water layer = $A \times 1 = A \text{ cm}^3$

Mass of water evaporating in the time.

$$T = A \times 1 = Ag$$

Amount of heat required to evaporate A g of water, i.e.,

$$Q = AL = A \times 540 \text{ [as L, latent heat of steam = 540 cal/g]}$$

$$= 540 A \text{ Cal} \quad \dots\dots(i)$$

Let T_1 be the temperature of the furnace and T_2 that of boiling water. Therefore,

$$T_1 - T_2 = (T_1 - 100)$$

Thickness of the tank, $x = 2 \text{ cm}$

$$\text{As } Q = \frac{kA(T_1 - T_2)t}{x} \quad \dots\dots(ii)$$

From eqns. (i) and (ii)

$$\frac{kA(T_1 - T_2)t}{x} = 540A \text{ or}$$

$$(T_1 - T_2) = \frac{540Ax}{kAt} = \frac{540x}{kt}$$

$$\text{or } T_1 - 100 = \frac{540 \times 2}{0.2 \times 540} = 10 \text{ or } T_1 = 100 + 10 = 110^\circ\text{C}$$

28. 00008.00

Sol. $(V_L)_{\text{rms}} = \frac{8}{\sqrt{2}} = 4\sqrt{2}$

$$(V_L)_{\text{peak}} = 8$$

29. 00005.00

Sol. Energy of gas due to thermal motion is transitional and rotational

$$\text{Since } P = \frac{2}{3} \left(\frac{\text{TKE}}{V} \right)$$

$$\Rightarrow \text{TKE} = \frac{3}{2} PV = \frac{3}{2} \frac{P_m}{\rho} = \frac{3}{2} \times \frac{8 \times 10^4 \times 1}{4} = 3 \times 10^4 \text{ J}$$

$$\therefore \text{RKE} = 2 \times 10^4 \text{ J}$$

$$\therefore \text{Total energy} = 5 \times 10^4 \text{ J}$$

30. 00000.50

Sol. Conceptual

Chemistry**PART – B****SECTION – A**

31. A

Sol. ${}_{56}\text{Ba}$ is a s-Block element.

32. C

Sol. CdS is yellow in colour.

33. D

Sol. The ring attached with nitrogen has more electron density, so Br^+ will come of para position of it.

34. D

Sol. H^- prefer direct attack on conjugated aldehyde.

35. B

Sol. $\begin{array}{c} \text{OH} \\ | \\ -\text{CH}-\text{CH}_3 \end{array}$ group will show haloform reaction.

36. C

Sol. $\Delta x \cdot \Delta p \geq \frac{h}{4\pi}$

$$\frac{1}{4} \Delta p \times \Delta p \geq \frac{h}{4\pi}$$

$$\Delta p \geq \sqrt{\frac{h}{\pi}}$$

$$\Delta v \geq \frac{1}{m} \sqrt{\frac{h}{\pi}}$$

37. D

Sol. When $x = 49$ ml, maximum heat will be evolved as maximum neutralization occurs.

38. D

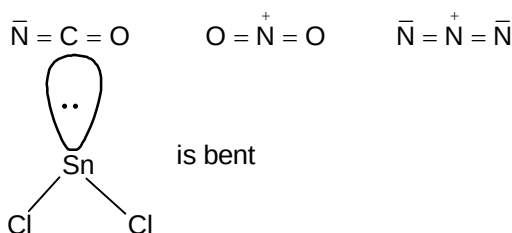
Sol. $\text{Rate} = -\frac{1}{2} \frac{d[\text{B}]}{dt} = \frac{1}{4} \frac{d[\text{C}]}{dt}$

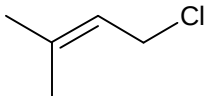
$$-\frac{d[\text{B}]}{dt} = \frac{1}{2} \frac{d[\text{C}]}{dt} = K[\text{A}]^2 [\text{B}]$$

$$\therefore \frac{d[\text{C}]}{dt} = 2K[\text{A}]^2 [\text{B}]$$

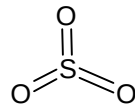
39. C

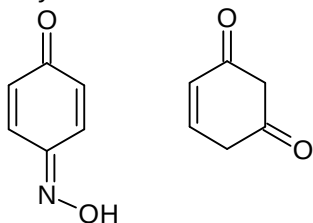
Sol. II, III & IV are linear



40. A
Sol.  will form most stable cation (allylic cation with tertiary at one end)
41. C
Sol. Common ion effect is observed in a mixture of strong and weak electrolyte having common ion.
42. A
Sol. Order of stability of conjugate base
 $\text{CH}_3\text{COO}^- > \text{CN}^- > \text{OH}^- > \text{C}_2\text{H}_5\text{O}^-$
43. C
Sol. Average kinetic energy of ideal gas is the function of temperature only.
44. D
Sol. As_2S_3 is a negatively charged sol, therefore ion with higher positive charge has higher coagulating power
45. B
Sol. $\text{CaC}_2 + \text{N}_2 \xrightarrow{\Delta} \text{CaCN}_2 + \text{C}$
46. C
Sol. Basic strength of alkali metal oxide increases down the group
47. B
Sol. Conceptual
48. B
Sol. Bauxite is an ore of Al
49. D
Sol. Analgesic drug is used to reduce the pain
50. A
Sol. V_2O_5 is used as catalyst for preparation of H_2SO_4 in contact process.

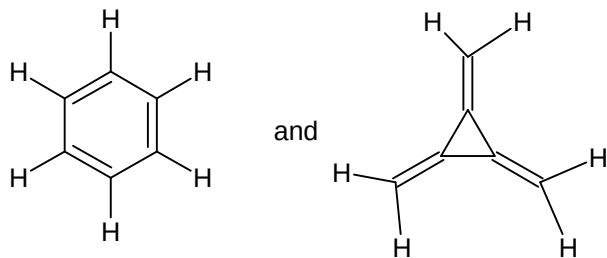
SECTION – B

51. 00001.50
Sol. 
 $\sigma\text{-bond} = 3$
 $p\pi - d\pi = 2$
 $p\pi - p\pi = 1$
52. 00002.00
Sol. Only these two will show tautomerism



53. 00002.00

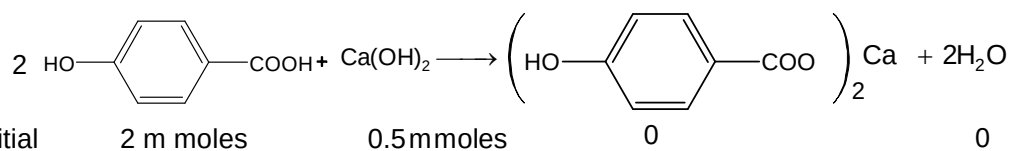
Sol.



Has 6-H equivalent.

54. 00005.00

Sol.



This is a buffer solution

pH = pKa + log $\frac{[\text{conjugate base}]}{[\text{acid}]}$

$$\text{pH} = \text{pKa} + \log \frac{[\text{conjugate base}]}{[\text{acid}]}$$

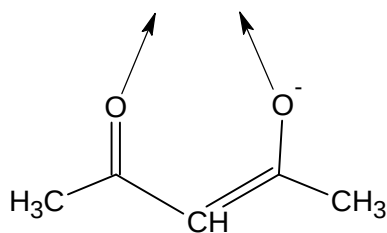
$$= 5 + \log \frac{1}{1}$$

$$= \text{pH} = 5$$

55. 00002.00

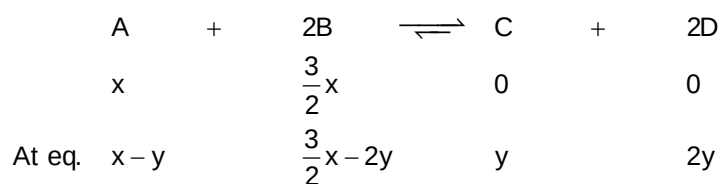
Sol.

Acetylacetonato anion is bidentate ligand, so two moles are needed



56. 00004.00

Sol.



According to question at equilibrium

$$x - y = \frac{3}{2}x - 2y$$

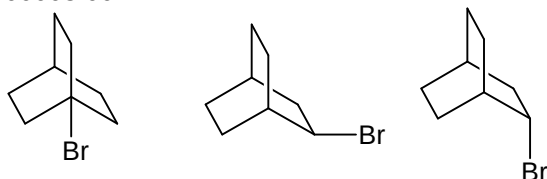
$$y = \frac{x}{2}$$

$$K_p = \frac{y \times (2y)^2}{(x - y) \left(\frac{3}{2}x - 2y \right)}$$

$$= \frac{\frac{x}{2} \times x^2}{\left(x - \frac{x}{2}\right) \left(\frac{3}{2}x - x\right)^2} = 4$$

57. 00003.00

Sol.



58. 00054.00

Sol.

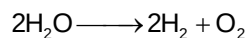
P. E of ccp is 74%

After removing half of the face centre of atom

$$\% \text{ P.E} = \frac{4 - \frac{3}{2}}{4} \times 74 = 46.25$$

$$\therefore \text{ void efficiency} = 100 - 46.25 = 53.75 \approx 54 \%$$

59. 00053.61

 Sol. Number of mole of $\text{H}_2\text{O} = \frac{36}{18} = 2$

 Number of e^- involved = 4

$$4 \times 96500 = 2 \times \text{time}$$

$$\text{time} = 2 \times 96500 \text{ sec} = 1.93 \times 10^5 \text{ sec} = 53.61 \text{ hrs}$$

60. 00030.00

 Sol. $P_{\text{final}} = \sqrt{P_x^0 \times P_y^0}$
 $= 30 \text{ mm}$

Mathematics**PART – C****SECTION – A**

61. B

Sol. If $|z_1| < 1 \Rightarrow |z_1|^5 + |z_1|^3 + |z_1| < 3$
 So, contradiction $3 < 3$

62. D

Sol. $t = e^{|\sin x|}$, $t \in [1, e]$

63. C

Sol. $2ae = 13$, $a^2(e^2 - 1) = 36$

64. A

Sol. $t \in \left(-\frac{3}{2}, -1\right) \cup \left(\frac{1}{2}, 1\right)$

65. A

Sol. $10 \int_{-\frac{1}{2}}^0 \ln(x+1) dx = 5 \ln\left(\frac{e}{2}\right)$

66. D

Sol. $6^4 - (4^4 - 3^4) = 1121$

67. A

Sol. $\frac{b}{2a} > 2$, $b^2 \geq 48a$, $2a - b + 12 > 0$
 $\Rightarrow (a, b) \equiv (1, 7)$

68. C

Sol. $x = 1$, $y = 1$, $f(1) = -2$, 3
 $f(x) = \frac{2(2x+1)}{x(f(1)-1)} \Rightarrow f(1) = -2$, $f\left(\frac{1}{2}\right) = -\frac{8}{3}$

69. D

Sol. $\min((\sin \alpha)x^2 - 2x - b - 2)$ must be greater than zero but minimum is at $x = 1$
 $\Rightarrow b \geq 4$

70. B

Sol. ${}^7C_2 = 21$

71. D

Sol. $L = \frac{1}{2}$

72. B

Sol. $-\frac{a}{b} = -1 \Rightarrow a = b$, $5a + 15b = 40 \Rightarrow a = b = 2$

73. B

Sol. $g(x) = 2 \sin^2\left(\frac{x}{2}\right) \leq \frac{x^2}{2}$

74. B

Sol. $(z-1)^5 = z^2(z-1) = 0 \Rightarrow (z-1)(z^2-3z+1)(z^2-z+1) = 0$

75. B

Sol. By the graph number of point of intersection will be 3

76. D

Sol. $A = \sum_{r=1}^{\infty} \frac{1}{r^2}, B = \sum_{r=1}^{\infty} \frac{1}{(2r-1)^2}$

77. D

Sol. $P(E) = \frac{{}^{13}C_4}{{}^{13}C_2 + {}^{35}C_2 + {}^{13}C_3 + {}^{39}C_1 + {}^{13}C_4}$

78. A

Sol. Coefficient of x^4 in $(1-x)^4(1-x^5)(1-x^6)(1-x)^{-3} = {}^6C_4 - 1 = 14$

79. B

Sol. Common point of contact is (0, 4)

80. A

Sol. Either $(\alpha-5)^2 > 4$ and $(\alpha+1)^2 = 4$ or $(\alpha-5)^2 = 4$ and $(\alpha+1)^2 > 4$

SECTION – B

81. 00000.14

Sol. $\alpha^2, \beta^2, \gamma^2$ are roots of the equation $t^3 - 21t^2 + 35t - 7 = 0$

82. 00004.00

Sol. $\operatorname{Re}(z) \leq |z|$

83. 00001.83

Sol. $b = \frac{5}{12}a - 1 \Rightarrow a = 2, b = -\frac{1}{6}$

84. 00000.75

Sol. $k = 0$

85. 00000.40

Sol. $p = (40)^4 - 5(40)^2 + 4$

86. 00001.50

Sol. $L_2 \equiv 2x - 3y = 0$

87. 00000.02

Sol. $-a + b - c + d = 9 \Rightarrow (9-a) + b + (9-c) + d = 9$ so number of such $P(x) = {}^{12}C_3$

88. 00001.98

Sol. $m = \frac{50}{99}$

89. 00002.00

Sol. $L = \frac{1}{2}, n = 4$

90. 00003.20

Sol. $n(S) = 2^6 \Rightarrow P = \frac{20}{2^6} = \frac{5}{16} \Rightarrow n(E) = 20$